

$$\|u-v\|^2 = \langle u+(-v), u+(-v) \rangle \stackrel{SL2}{=} \langle u+(-v), u \rangle - \langle u+(-v), v \rangle \stackrel{(3.2)}{=} \langle u, u \rangle - \langle v, u \rangle - \langle u, v \rangle + \langle v, v \rangle$$

$$\stackrel{(SSS)}{=} \langle u, u \rangle - \overline{\langle u, v \rangle} - \langle u, v \rangle + \langle v, v \rangle$$

$$\|u+v\|^2 - \|u-v\|^2 = \cancel{\|u\|^2} + \cancel{\|v\|^2} + 2\operatorname{Re}(\langle u, v \rangle) = \|u\|^2 + \|v\|^2 - 2\operatorname{Re}(\langle u, v \rangle)$$

$$= \cancel{\|u\|^2} - \cancel{\|u\|^2} + 2\operatorname{Re}(\langle u, v \rangle)$$

$$\|u+v\|^2 - \|u-v\|^2 = 4\operatorname{Re}(\langle u, v \rangle) \quad \therefore 4$$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{\|u+v\|^2 - \|u-v\|^2}{4}$$